The Category of Persistent Observational-Transfer Objects

A Proposed Paper AA for the Dream-Paper Suite

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Abstract

The dream-paper suite has, as its ultimate outer-envelope target, a Unified Emergence Theorem: the existence of a unique minimal persistent object $\mathcal{U}$ from which every higher physical layer — persistent sector structure, matter carriers, causal order, entropy production, continuum geometry, and field dynamics — arises as a derived realization shadow, not as an independently inserted primitive.

Before that theorem can be stated with mathematical precision, its home category must be defined. The present paper introduces **POT**, the *category of persistent observational-transfer objects*, which is the native mathematical environment in which $\mathcal{U}$ lives. An object of **POT** is a five-tuple $(\Sigma, \Lambda, \mathsf{T}, \mathsf{R}, \mathsf{D})$ encoding a stabilized observational quotient, a continuation-enlargement structure, an admissible transfer law, an optional representation layer (populated only when licensed by bridge theorems), and a defect-ledger. **POT** is constructed as a fibred category over a base category **Reg** of admissible regimes and enlargements.

The paper has three main contributions. First, it defines the objects, morphisms, and fibration structure of **POT** in terms of data already licensed by the finite kernel and companion papers. Second, it states the Overlap Route-Agreement Lemma as the first theorem whose proof would make $\mathcal{U}$'s existence mathematically unavoidable rather than aspirational. Third, it maps the theorem ladder — from route agreement to coherent transition maps to the transport invariant and its uniqueness — that the suite must climb before the universal-completion construction of $\mathcal{U}$ becomes viable.

No claim in this paper exceeds the status of the upstream stack it imports. **POT** is a definition paper together with a named-obligation map; it does not itself prove the Unified Emergence Theorem or any major fragment thereof.

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1 Purpose and imports

1.1 Why a new paper is needed here

The bridge chain

$$E \to M \to N \to O \to P \to Q \to R \to S \to T \to U \to V \to W \to X \to Y \to Z$$

produces, under explicit hypotheses, a limit-domain Yang–Mills closure theorem. Papers Y and Z use the phrase "universal continuum realization" and appeal to a colimit construction in an unspecified "compatible limit topology category" $\mathcal{C}_{\mathrm{lim}}$.

That category is not yet defined inside the suite. Its absence means the language of Papers Q–Z is partially anticipatory: the objects being assembled are named and their properties are assumed, but the ambient mathematical environment in which they live is left implicit.

The present paper remedies this. It defines the category **POT** — the smallest mathematically precise home for the objects that the suite's bridge papers have been constructing. Once **POT** is in place:

1. the phrase "compatible limit topology" in Paper Z's Hypothesis $\mathrm{H}_{\mathrm{LDA}}$ acquires a precise referent;
2. the universal object $\mathcal{U}$ of the Unified Emergence Theorem has a well-defined home;
3. the theorem ladder from the NFC Push program — Overlap Route-Agreement, Coherent Transition Maps, Transport Invariant, Minimal Carrier — has a precise mathematical setting.

1.2 What POT imports

POT is built exclusively from ingredients already licensed by the upstream suite:

1. the finite kernel's observational quotient Σ, stabilization, and defect-ledger structure (Paper A);
2. the admissible-test and continuation discipline, including the lifted survivor space $\widetilde{\Sigma} = \Sigma \times \mathbb{N}$ with cumulative defect count $\Lambda(h)$ (flagship monograph);
3. the LS-2 boundary law and realized stabilized collar/outcome data (Paper B);
4. the certificate doctrine forbidding regime enlargement without an explicit transfer theorem (Paper D);
5. the sector-extraction program's explicit demand for persistence, coherent transition maps, canonical projection, and continuum-relevant interpretation (Paper E); and
6. the colimit language anticipated but not grounded by Papers Q–Z.

1.3 What POT does not import

POT is defined *before* any of the following are proved, and must not presuppose:

(i) persistence of any extracted sector (Paper E, open);

(ii) canonical projection to any target factor (Paper E, open);

(iii) a continuum realization of any persistent object (O-CONT, open);

(iv) Yang–Mills structure, spectral separation, or mass-gap transfer; or

(v) any matter representation, causal, or gravitational layer.

These are downstream realizations of **POT**-objects, not ingredients of **POT** itself.

2 The base category of admissible regimes

Definition 2.1 (Admissible regime)**.** An *admissible regime* is a finite relational configuration space $R = (\Omega_R, \mathcal{A}_R)$ consisting of a finite universe Ω_R together with a collection $\mathcal{A}_R$ of admissible tests on Ω_R in the sense of the flagship kernel (Paper A).

Definition 2.2 (Admissible enlargement)**.** An *admissible enlargement* $\iota : R \hookrightarrow R'$ is an injective map $\Omega_R \hookrightarrow \Omega_{R'}$ that is compatible with admissible tests in the sense of Paper D: the enlarged regime introduces no new primitive data forbidden by the certificate doctrine.

Definition 2.3 (Base category **Reg**)**.** The *base category of admissible regimes* **Reg** has:

- *objects*: admissible regimes R;
- *morphisms*: admissible enlargements $\iota : R \hookrightarrow R'$; and
- *composition*: composition of injective enlargement maps, which is again admissible.

Reg is a small category.

Remark 2.4 (Why **Reg** is not just a poset). Admissible enlargements are not merely inclusions ordered by containment. Different enlargement paths $R \hookrightarrow R' \hookrightarrow R''$ and $R \hookrightarrow R'' \hookrightarrow R'''$ may produce different intermediate regimes R' and R'''. The category structure of **Reg** keeps track of this path information, which is exactly what is needed to state the Overlap Route-Agreement Lemma in Section 7.

3 The fibers of POT

Over each regime $R \in \mathbf{Reg}$, the fiber $\mathcal{F}_R$ records all the licensed observational-transfer data associated to R.

Definition 3.1 (Fiber over a regime)**.** The *fiber* $\mathcal{F}_R$ over an admissible regime R consists of the following data, all licensed by the upstream suite:

1. **Stabilized observational quotient** Σ_R: the behavioral congruence quotient of Ω_R by the admissible tests $\mathcal{A}_R$, as defined in Paper A. Its cardinality $|\Sigma_R|$ is the *observable size* of the regime.

2. **Continuation-enlargement structure** Λ_R: the continuation data on R, including continuation histories h, cumulative defect count $\Lambda(h)$ in the lifted survivor space $\widetilde{\Sigma}_R = \Sigma_R \times \mathbb{N}$, and the right-stable / defect-mediated dichotomy from the flagship monograph.

3. **Transfer structure** T_R: the admissible transfer maps on R, including the boundary-law map $\Phi_R : C_{\rho^*}(R) \to O(R)$ from realized stabilized collar types to realized stabilized local outcomes, as specified under LS-2 (Paper B).

4. **Representation layer** R_R: optional. Populated only when a bridge theorem licenses it. Initially empty for all R.

5. **Defect-ledger data** D_R: the defect-ledger recording, for each continuation history h on R, the cumulative defect count $\Lambda(h)$ and its decomposition into right-stable entropy drops and defect-mediated increments.

Remark 3.2 (On the optional representation layer). Item (4) in Definition 3.1 is the suite's anti-smuggling firewall at the categorical level. The fiber is defined without assuming sector persistence, canonical projection, or continuum realization. Those features enter only when the corresponding bridge theorems — Paper E's persistence/projection program, O-CONT, Paper F's Yang–Mills realization — are proved. At that point the representation layer of the relevant fiber is populated by a downstream functor, not retroactively assumed.

4 Objects and morphisms of POT

Definition 4.1 (Object of **POT**). A *persistent observational-transfer object* is a pair $(R, \mathcal{F}_R)$ where R is an admissible regime and $\mathcal{F}_R$ is its fiber as in Definition 3.1. We write objects as $\mathcal{X}_R$ when the regime is relevant, and suppress R when it is clear.

Definition 4.2 (Morphism of **POT**). A **POT**-*morphism* $f : \mathcal{X}_R \to \mathcal{X}'_{R'}$ over an admissible enlargement $\iota : R \hookrightarrow R'$ consists of structure-preserving maps between the fiber components:

1. $f_\Sigma : \Sigma_R \to \Sigma_{R'}$ compatible with the observational quotient structure;

2. $f_\Lambda : \Lambda_R \to \Lambda_{R'}$ compatible with continuation histories and defect-ledger data, in the sense that $\Lambda_{R'}(f_\Lambda(h)) = \Lambda_R(h) + \delta_\iota(h)$ for an explicit defect increment $\delta_\iota(h) \geq 0$;

3. $f_T : \mathsf{T}_R \to \mathsf{T}_{R'}$ compatible with the boundary-law maps and the LS-2 collar/outcome discipline; and

4. $f_D : \mathsf{D}_R \to \mathsf{D}_{R'}$ consistent with defect-ledger additivity.

The representation layer, when non-empty, must also be transported compatibly. A morphism may transport an empty representation layer to another empty one; populating it requires a bridge theorem.

Proposition 4.3 (Composition of morphisms). **POT**-*morphisms compose: if* $f : \mathcal{X}_R \to \mathcal{X}'_{R'}$ *lies over* $\iota : R \hookrightarrow R'$ *and* $g : \mathcal{X}'_{R'} \to \mathcal{X}''_{R''}$ *lies over* $\kappa : R' \hookrightarrow R''$*, then* $g \circ f$ *lies over* $\kappa \circ \iota$ *and is a valid* **POT**-*morphism.*

Proof. Composition of the component maps $f_\Sigma, f_\Lambda, f_T, f_D$ is straightforward. The defect-ledger additivity requires that $\delta_{\kappa\circ\iota}(h) = \delta_\kappa(f_\Lambda(h)) + \delta_\iota(h)$, which holds by the additive structure of the lifted survivor space. □

Definition 4.4 (The category **POT**)**.** The *category of persistent observational-transfer objects* **POT** has:

- *objects*: persistent observational-transfer objects $(R, \mathcal{F}_R)$;
- *morphisms*: **POT**-morphisms as above; and
- *forgetful functor* $\pi : \mathbf{POT} \to \mathbf{Reg}$ sending $(R, \mathcal{F}_R) \mapsto R$.

POT is a category fibred over **Reg**.

Remark 4.5 (Why **POT** is richer than a presheaf category). A presheaf on **Reg** would assign a set to each regime and a restriction map to each enlargement. **POT**-fibers are not merely sets; they carry quotient structure, defect-ledger data, and boundary-law maps. This extra structure is what allows the transport invariant $I(R) = \log|\Sigma_R| + \lambda\Lambda_R$ to be defined as a **POT**-native quantity rather than an external assignment.

5 Reindexing and the fibration structure

Definition 5.1 (Reindexing functor)**.** For each admissible enlargement $\iota : R \hookrightarrow R'$, the *reindexing functor* $\phi^*_{R\to R'} : \mathcal{F}_{R'} \to \mathcal{F}_R$ restricts the data of R' back to R. Explicitly:

1. the observational quotient restricts along ι;
2. continuation histories on R' that originate in R restrict to continuation histories on R;
3. the boundary-law map restricts to the boundary-law map on R; and
4. the defect ledger restricts by forgetting increments introduced by ι.

Proposition 5.2 (Cartesian morphisms)**.** *Every* **POT**-*morphism over an admissible enlargement* ι *admits a Cartesian lift, making* $\pi : \mathbf{POT} \to \mathbf{Reg}$ *a Grothendieck fibration.*

Proof. The Cartesian lift of $\iota : R \hookrightarrow R'$ at a fiber object in $\mathcal{F}_R$ is the morphism induced by the reindexing functor $\phi^*_{R\to R'}$. The universal property of Cartesian morphisms follows from the fact that the fiber data (quotient, ledger, transfer) are defined canonically from the regime data, with no ambiguity. □

6 The transport invariant as a POT-native quantity

With the category **POT** in place, the transport invariant proposed in the NFC Push program acquires a precise categorical home.

Definition 6.1 (Compression-ledger transport invariant)**.** For an object $(R, \mathcal{F}_R) \in \mathbf{POT}$ and a local region $U \subseteq \Omega_R$, the *compression-ledger transport invariant* is

$$I(R, U) \;=\; \log|O_R(U)| \;+\; \lambda\,\Lambda_R(U) \;+\; \mu\,\log|C_R(U)|$$

where $O_R(U)$ is the set of realized stabilized local outcomes, $C_R(U) = C_{\rho^*}(U)$ is the set of realized stabilized collar types under LS-2, $\Lambda_R(U)$ is the cumulative defect-ledger count transported to U, and $\lambda, \mu \geq 0$ are normalization coefficients to be determined by the transport law.

The *boundary-corrected nonlinear extension* is:

$$I_{\mathrm{nl}}(R, U) \;=\; \log|O_R(U)| \;+\; \lambda\,\Lambda_R(U) \;+\; \mu\,\log|C_R(U)| \;+\; \nu\,\Lambda_R(U)\cdot\log|C_R(U)|$$

with $\nu \geq 0$ an additional interaction coefficient.

Remark 6.2 (On the naturality of I). The quantities $|O_R(U)|$ and $|C_R(U)|$ are defined at the level of realized stabilized data, not raw token counts. This is the suite's core operational discipline: only what survives admissible tests and stabilization is licensed as objective. The defect count $\Lambda_R(U)$ lives in the continuation-enlargement structure Λ_R. So $I(R, U)$ is a function of **POT**-native data and is well-defined without appealing to any downstream bridge content.

7 The Overlap Route-Agreement Lemma

With **POT** defined, the Overlap Route-Agreement Lemma can be stated in precise categorical terms. It is the first theorem whose proof would make the construction of a universal object $\mathcal{U} \in \mathbf{POT}$ mathematically unavoidable.

Definition 7.1 (Compatible overlap)**.** Two admissible enlargements $\iota_1 : Q \hookrightarrow R$ and $\iota_2 : Q \hookrightarrow R'$ sharing a common source Q define a *compatible overlap* if there exists an admissible regime R'' and enlargements $\kappa_1 : R \hookrightarrow R''$ and $\kappa_2 : R' \hookrightarrow R''$ such that $\kappa_1 \circ \iota_1 = \kappa_2 \circ \iota_2$ in **Reg**.

Lemma 7.2 (Overlap Route-Agreement Lemma — target statement)**.** *Let $\iota_1 : Q \hookrightarrow R$ and $\iota_2 : Q \hookrightarrow R'$ be two admissible enlargements with a compatible overlap completing to R''. Let E_R and $E_{R'}$ be two licensed extraction procedures on R and R' respectively, producing stabilized observational-transfer data on each. Then the restrictions of E_R and $E_{R'}$ to Q are canonically equivalent: there exists a unique* **POT***-isomorphism*

$$\theta_{R,R';Q} : E_R|_Q \xrightarrow{\simeq} E_{R'}|_Q$$

compatible with all fiber components $(\Sigma, \Lambda, \mathsf{T}, \mathsf{D})$.

Remark 7.3 (Status of Lemma 7.2). Lemma 7.2 is a *target theorem*, not a present theorem. Its status in the master ledger is Open. The present paper states it precisely for the first time so that the suite knows exactly what it is trying to prove. The NFC Push program identifies this as the "first forcing theorem": without it, the phrase "universal persistent object" is aspirational; with it, the phrase becomes a legitimate construction problem.

Remark 7.4 (Why this is the correct first theorem). The Overlap Route-Agreement Lemma is the descent seed for the **POT**-fibration. A **POT**-object on R restricts to one on Q via the reindexing functor; the lemma says two different **POT**-objects on R and R' agree on their shared source Q up to canonical equivalence. This is the weakest honest statement of local coherence, and it is the precise condition needed to begin assembling a global persistence diagram from local fiber data.

8 The theorem ladder to $\mathcal{U}$

The Push program identifies a six-step ladder from the Overlap Route-Agreement Lemma to the existence of the universal object $\mathcal{U}$. Each step is a named open obligation in the master ledger. The present paper records them as a dependency chain.

Step 1. **Overlap Route-Agreement** (Lemma 7.2): Licensed extraction procedures agree on overlaps up to canonical **POT**-equivalence. This is the descent seed.

Step 2. **Coherent Transition-Map Theorem**: For every admissible enlargement $\iota : R \hookrightarrow R'$ there exists a canonical structure-preserving **POT**-morphism $\tau_\iota : \mathcal{F}_R \to \mathcal{F}_{R'}$, satisfying identity and composition axioms and agreeing with the overlap equivalences of Step 1. This converts the set of fiber data into a genuine directed diagram in **POT**.

Step 3. Compression-Ledger Transport Theorem: The invariant $I(R,U)$ of Definition 6.1 is witness-descending, extension-stable on right-stable enlargements, and continuation-covariant. The normalization coefficient λ is uniquely determined by the composition-additivity condition on the coherent transition maps.

Step 4. Uniqueness of (λ, μ, ν): The full invariant $I_{\mathrm{nl}}(R,U)$ has unique normalization coefficients fixed by route-invariance (using Step 1), composition-additivity (using Step 2), and collar-context consistency (using the LS-2 data in the fiber).

Step 5. Amplified Minimal-Carrier Theorem: Under the corrected transport invariant, there exists at most one minimal enlargement-compatible carrier family $\mathcal{K}^*$ through which I_{nl} factors, with any residual ambiguity isolated as a later projection obstruction. This is the first theorem that selects a genuine minimal persistent carrier inside **POT**.

Step 6. Universal Completion of the Persistence Diagram: The directed diagram of **POT**-objects and coherent transition maps (Steps 1–2) admits a universal completion $\mathcal{U} \in \mathbf{POT}$ satisfying:

(a) every stabilized admissible observable factors through $\mathcal{U}$;

(b) every enlargement-compatible presentation maps coherently into $\mathcal{U}$;

(c) $\mathcal{U}$ is minimal among persistent objects with property (a);

(d) every faithful realization into an algebraic, operator, geometric, or other licensed bridge category is determined by the realization of $\mathcal{U}$;

(e) $\mathcal{U}$ is unique up to unique equivalence.

Remark 8.1 (On the bi-universal property of $\mathcal{U}$). The universal property in Step 6 is *bi-universal*: $\mathcal{U}$ is terminal for lawful observable factoring and initial for faithful licensed realizations. It is not an initial or terminal object of **POT** in the naive sense; it is a reflective core satisfying both directions simultaneously. This structure is necessary because the suite has two distinct directions of interest: collecting all stabilized observables (terminal direction) and generating all faithful realizations (initial direction).

9 Realization functors out of POT

Once $\mathcal{U}$ is constructed, all subsequent bridge realizations should be expressed as functors out of **POT**. The following taxonomy maps the suite's existing bridge papers to their **POT**-realization roles.

Definition 9.1 (Realization functor)**.** A *realization functor* is a functor $F : \mathbf{POT} \to \mathcal{C}$ to a licensed bridge category $\mathcal{C}$ that:

1. preserves the **POT**-fibration structure;
2. maps **POT**-morphisms to morphisms of $\mathcal{C}$; and
3. is licensed by an explicit bridge theorem in the suite.

Suite paper	Realization category $\mathcal{C}$	What the functor extracts
Paper E	Finite algebraic sector category	Persistent sector data from R
Papers N–P	Operator / semigroup category	Profile, carrier, generator from T
Papers Q–R	Continuum realization category	Universal limit of $\mathcal{U}$
Paper S	Yang–Mills structure category	Gauge interpretation of realized object
Papers T–Z	Spectral / domain-closure chain	Spectral transfer and exhaustion
Future matter paper	Representation category over gauge sector	Canonical matter carriers
Future causal paper	Causal preorder category	Transfer-order reconstruction

Remark 9.2 (Representation layer population via realization functors). The empty representation layer R_R in Definition 3.1 is populated precisely when a realization functor $F : \mathbf{POT} \to \mathcal{C}$ is applied and the bridge theorem governing F licenses a non-trivial assignment. This is the **POT**-level implementation of Paper D's anti-smuggling doctrine: representation content cannot be pre-loaded into the fiber; it must arrive via an explicit licensed functor.

10 Relationship to Papers Q–Z

Papers Q–Z use the language of "compatible limit topology" and "colimit in $\mathcal{C}_{\mathrm{lim}}$" without specifying the home category. **POT** supplies that home category.

Specifically:

1. Paper Z's Hypothesis $\mathrm{H}_{\mathrm{LDA}}$ (limit-domain assembly) is now precisely the statement that the directed diagram of **POT**-objects along the exhaustion $(\mathcal{D}_n)$ admits a colimit in **POT**.

2. Paper Z's Hypothesis $\mathrm{H}_{\mathrm{UGSC}}$ (uniform gap-subcriticality) is an additional quantitative condition on the spectral-transfer realization functor applied to the colimit object.

3. The limit domain $\mathcal{D}_\infty = \bigcup_n \mathcal{D}_n$ corresponds to the image of the colimit **POT**-object under the continuum realization functor $F : \mathbf{POT} \to \mathcal{C}_{\mathrm{cont}}$.

This retro-grounding does not change any theorem in Papers Q–Z. It clarifies their categorical setting and makes their hypotheses checkable at the right level of abstraction.

11 Status ledger

Label	Claim	Status
AA1	Definition of **POT** as a Grothendieck fibration over **Reg**	UNC (definition)
AA2	Definition of the transport invariant $I(R,U)$ and I_{nl}	UNC (definition)
AA3	Overlap Route-Agreement Lemma	OPEN
AA4	Coherent Transition-Map Theorem	OPEN
AA5	Uniqueness of normalization (λ, μ, ν)	OPEN
AA6	Amplified Minimal-Carrier Theorem	OPEN
AA7	Universal Completion: existence of $\mathcal{U} \in \mathbf{POT}$	OPEN
AA8	Bi-universal property of $\mathcal{U}$	OPEN
AA9	Unified Emergence Theorem	OPEN / far downstream
AA10	Retro-grounding of Papers Q–Z's colimit language in **POT**	COND (on AA7)

12 Conclusion

Paper AA introduces the category **POT** of persistent observational-transfer objects as the natural mathematical home for the suite's outer-envelope ambitions. **POT** is a Grothendieck fibration over the base category **Reg** of admissible regimes and enlargements. Its fibers carry only data already licensed by the finite kernel: stabilized observational quotients, continuation-enlargement structure with defect ledger, LS-2 boundary-law transfer data, and an empty representation layer that is populated only by explicit downstream bridge functors.

The category **POT** serves three roles simultaneously. First, it gives the colimit language of Papers Q–Z a precise mathematical home. Second, it provides the setting in which the transport invariant $I(R,U)$ and its nonlinear extension I_{nl} are **POT**-native quantities rather than ad hoc formulas. Third, it makes the Unified Emergence Theorem mathematically precise: the universal object $\mathcal{U} \in \mathbf{POT}$, if it exists, is the unique minimal persistent object that is terminal for observable factoring and initial for faithful licensed realizations, with all bridge papers of the suite providing realization functors out of **POT**.

The Overlap Route-Agreement Lemma (AA3) is the first theorem whose proof would make the construction of $\mathcal{U}$ mathematically unavoidable. Until that lemma is proved, $\mathcal{U}$ is a precisely stated aspiration. After it is proved, the theorem ladder of Section 7 converts it into a legitimate construction program.

That is the correct posture for this paper. It does not prove the Unified Emergence Theorem. It builds the room in which that theorem will live, identifies the door that must be opened first, and names the keys in their correct order.